CIVICEYE: CROWDSOURCED CIVIC ISSUE REPORTING AND RESOLUTION SYSTEM

PROJECT REPORT

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ABSTRACT

Civic issues such as damaged roads, water supply problems, waste accumulation, and sanitation deficiencies affect public safety and community well-being. Traditional grievance management often faces challenges involving manual complaint handling, inefficient departmental coordination, duplicate submissions, and limited transparency. To address these issues, CivicEye: An AI-Assisted Civic Grievance Management Platform provides a centralized digital solution for complaint reporting, processing, assignment, and resolution tracking. The platform connects citizens and local authorities through role-based interfaces for Citizens, Department Heads, Verifiers, and Super Administrators. Citizens can submit complaints with descriptions and image evidence, specify locations using an interactive map, and track complaint status. Department Heads manage complaints and assign Verifiers, while Verifiers inspect assigned issues and update their progress. Super Administrators monitor platform activity through analytical dashboards. CivicEye integrates AI-assisted complaint classification, severity assessment, semantic duplicate detection, and geospatial processing to support efficient grievance management. Authentication and role-based access control help regulate access to system functionalities. By combining intelligent processing, location-aware reporting, and administrative analytics, CivicEye aims to improve transparency and coordination between citizens and authorities, with future potential for integration into municipal systems.

Keywords: Civic Grievance Management, CivicEye, Artificial Intelligence, Complaint Classification, Semantic Duplicate Detection, Geospatial Analysis, Complaint Tracking, Role-Based Access Control, Departmental Management, Civic Analytics, Smart Governance.

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CHAPTER I

INTRODUCTION

1.1 Overview

Civic infrastructure such as roads, street lighting, water supply and waste collection is used by every citizen, yet the process of reporting a fault in it is frequently manual, scattered across channels and difficult to follow. **CivicEye: Crowdsourced Civic Issue Reporting and Resolution System** is a web-based platform that brings citizens, municipal departments, field verifiers and administrators into one digital workflow. A citizen reports an issue with a description, a location and supporting evidence; the platform checks the report for duplicates, classifies it into a civic category, estimates its severity and priority, and routes it to the responsible department, where a Department Head assigns it to a verifier for field inspection until it is resolved and confirmed by the citizen.

1.2 Background and Motivation

Online grievance redressal has been studied as part of e-government adoption, and the literature highlights that accessibility, citizen participation and usability decide whether such systems are actually used. Crowdsourced reporting platforms allow citizens to act as distributed sensors who describe problems as they see them, and centralized tracking gives authorities a single place to manage them. More recent work explores artificial intelligence to detect civic issues from geotagged visual evidence and to detect duplicate complaints, while sentence-embedding techniques provide a practical way to compare the meaning of two pieces of text.

The motivation for CivicEye comes from combining these ideas in a single, practical workflow: simple citizen reporting, location capture on an interactive map, semantic duplicate detection, AI-assisted classification and prioritization, department-specific work queues, verifier assignment and administrative analytics, all under role-based access control.

1.3 Problem Statement

Existing civic complaint systems often rely on manual reporting and fragmented communication between citizens and municipal departments. Duplicate complaints,

incorrect categorization, delayed assignment and limited visibility into complaint progress can reduce the efficiency of issue resolution.

1.4 Aim and Scope

1.4.1 Aim

To develop CivicEye, a centralized crowdsourced civic complaint management platform that enables citizens to report civic issues and helps municipal departments efficiently classify, assign, monitor and resolve them.

1.4.2 Scope

The scope of the project is defined by the responsibilities of four user roles: Citizen, Department Head, Verifier, and Super Admin. The Citizen is responsible for securely registering and authenticating, submitting civic complaints, uploading supporting evidence, providing location and complaint details, viewing personally submitted complaints, tracking complaint progress, and confirming resolution. The Department Head receives department-specific complaints, reviews complaint information, accesses citizen contact information when required, assigns complaints to department verifiers, and updates the official complaint status. The Verifier views complaints assigned by the Department Head, inspects reported civic issues, performs field verification, and reports the issue status to the Department Head. The Super Admin oversees all departments, monitors overall system progress, accesses department-wise analytics, and inspects system performance.

1.5 Methodological Overview

CivicEye follows a three-tier web architecture. The presentation layer is a React-based single-page frontend with role-specific dashboards. The application layer is a Node.js and Express REST API that implements authentication, complaint management, duplicate detection, classification, the priority engine, department assignment, analytics and notifications. The data layer is a MongoDB database (MongoDB Atlas) accessed through Mongoose. Intelligent components include semantic text embeddings with similarity comparison for duplicate detection, AI-assisted classification using Google Gemini against a centralized civic taxonomy, severity prediction, deterministic priority calculation and a rule-based fallback classifier. Complaint locations are stored as

GeoJSON points with a 2dsphere index so that nearby candidate complaints can be retrieved efficiently. The detailed methodology is presented in Chapter III.

1.6 Challenges

The existing solutions identifies several challenges in existing civic complaint handling that the CivicEye system is designed to address. These include manual or fragmented complaint reporting, repeated complaints for the same civic issue, incorrect or inconsistent complaint categorization, difficulty prioritizing critical issues, delayed assignment to responsible personnel, limited transparency for citizens, and the lack of centralized departmental monitoring. CivicEye aims to overcome these challenges by providing a centralized platform for complaint submission, duplicate detection, AI-assisted classification, priority assessment, and department-specific complaint assignment. The system also focuses on maintaining a simple and user-friendly experience for citizens while performing intelligent processing in the background. Furthermore, it ensures departmental data isolation through role-based access controls.

1.7 Expected Outcomes

CivicEye provides a unified digital workflow connecting Citizen → Department Head → Verifier → Resolution → Citizen Confirmation. The improvements expected from this workflow are faster complaint routing, reduced duplicate reports, better prioritization, clear responsibility, improved transparency and better departmental monitoring.

CHAPTER II

LITERATURE SURVEY

2.1 Introduction

This chapter covers online grievance redressal and e-government adoption, crowdsourced civic reporting, semantic text similarity, and AI-based geotagged civic issue detection. The information gathered from each work and its relevance to CivicEye is summarized in Table 2.1.

2.2 Existing Systems

Conventional civic complaint handling depends on manual channels, where a citizen informs an office and the complaint is recorded and forwarded by staff. As stated in the problem statement, such reporting is fragmented, offers limited visibility of progress to the citizen, and gives departments little consolidated information about repeated or critical issues. Digital grievance portals and crowdsourced reporting applications reduce some of this friction by capturing complaints electronically and routing them to the concerned authorities.

2.3 Digital Civic Grievance Platforms

The information gathered supports the need for an accessible digital civic grievance platform and highlights the importance of citizen participation and usability. This directly influences the CivicEye design goal of a simple citizen complaint submission flow with minimal training requirements.

The work titled “Crowdsourced Civic Issues Reporting and Resolution System” [3] developed a digital platform through which citizens report civic issues and complaints are routed to the concerned authorities. It establishes the need for crowdsourced civic issue reporting and supports centralized complaint tracking and authority-side management, which are both central to CivicEye.

2.4 Semantic Duplicate Detection

Semantic Duplicate Detection identifies multiple complaints that describe the same civic issue using different words or descriptions. Instead of relying only on exact keyword matching, the system analyzes the meaning and context of complaint text to

identify similar reports. For example, reports such as “large pothole near the bus stand” and “damaged road beside the bus stop” may refer to the same issue. By detecting potential duplicates, CivicEye can reduce redundant complaints, help authorities consolidate reports, and improve the efficiency of complaint tracking and resolution.

2.5 AI-assisted Complaint Classification

AI-Assisted Complaint Classification automatically categorizes citizen complaints based on their descriptions and, where applicable, supporting images. The system identifies the nature of the issue and assigns it to a suitable category, such as road damage, water supply, waste management, or sanitation. This reduces the need for manual classification and helps direct complaints to the appropriate department. In CivicEye, automated classification can improve complaint handling efficiency, support faster decision-making, and help authorities prioritize issues based on their urgency and potential public impact.

2.6 Geospatial Civic Issue Detection

Geospatial Civic Issue Detection uses geographical coordinates and location-based information to identify, map, and analyze civic problems across different areas. CivicEye can use the location attached to each complaint to display reported issues on an interactive map, identify areas with repeated problems, and highlight potential hotspots. For example, multiple reports of road damage within the same locality can help authorities identify roads that require inspection or maintenance. This location-based approach supports better resource allocation, targeted infrastructure maintenance, and data-driven planning for local authorities.

Table 2.1: Summary of Literature Reviewed

S.No.	Work	Work carried out	Information gathered relevant to CivicEye
1	Adoption of Online Public Grievance Redressal System in India – N. P. Rana, Y. K. Dwivedi, M. D. Williams and V. Weerakkody [1]	Studied adoption of online public grievance redressal systems; analysed citizen interaction with e-government platforms.	Supports the need for an accessible digital civic grievance platform; highlights the importance of citizen participation and usability.
2	Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks – N. Reimers and I. Gurevych [2]	Used cosine similarity to measure semantic similarity between texts.	Conceptual basis for semantic duplicate complaint detection.
3	Crowdsourced Civic Issues Reporting and Resolution System [3]	Developed a digital platform for citizens to report civic issues and routed complaints to concerned authorities.	Establishes the need for crowdsourced civic issue reporting; supports centralized complaint tracking and authority-side management.
4	Civi-Snap: A Deep Learning Powered Geotagged Civic Issue Detection and Monitoring Framework [4]	Explored AI-based civic issue detection; used geotagging and visual evidence; incorporated duplicate complaint detection.	Supports location-aware complaint processing; demonstrates the value of duplicate detection in civic systems.

Table 2.1 presents the summary of the literature reviewed for the CivicEye project. The reviewed studies focus on online complaint systems, duplicate complaint detection, civic issue reporting, and location-based monitoring. These studies help us understand how to improve complaint management and citizen participation. The findings support the development of CivicEye, which allows users to report civic issues, identify duplicate complaints, assign complaints to the concerned departments, and track their status until resolution.

2.7 Limitations of Existing Approaches

Based on the reviewed works and the problem statement, several limitations of existing approaches are identified that CivicEye aims to address. Complaint reporting and authority-side handling often remain fragmented, providing citizens with limited visibility into the progress of their complaints. In the absence of semantic comparison techniques, multiple reports describing the same civic issue may be treated as separate complaints, resulting in duplication of effort. Manual complaint categorization can also lead to inconsistencies and delays in routing complaints to the appropriate departments. Furthermore, the lack of a consistent prioritization method makes it difficult to identify and address critical issues promptly. Delays in assigning complaints to responsible personnel can occur when department-level workflows are not properly established. Additionally, the absence of centralized monitoring and analytical tools limits the ability of departments and administrators to track complaint status, evaluate performance, and manage civic issues efficiently. These limitations highlight the need for an integrated system such as CivicEye that supports intelligent complaint processing, efficient assignment, transparent tracking, and centralized administrative monitoring.

2.8 Proposed CivicEye Approach

CivicEye proposes a three-tier web platform in which a complaint passes through validation and duplicate detection, AI-assisted classification, priority calculation and department assignment before being handled by a Department Head and verifier, and finally confirmed by the citizen. The approach is described in detail in Chapters III and IV.

2.9 Summary

The literature supports the need for an accessible digital grievance platform, crowdsourced reporting with centralized tracking, semantic similarity for duplicate detection and location-aware processing. CivicEye combines these elements into one workflow.

CHAPTER III

OBJECTIVES AND METHODOLOGY

3.1 Introduction

The primary objective of CivicEye is to develop an intelligent and user-friendly platform that simplifies civic issue reporting and improves the efficiency of grievance resolution by local authorities. The system aims to automate complaint classification, identify potential duplicate reports, utilize geographical information to locate civic issues, and support urgency-based prioritization for effective resource allocation. The methodology involves gathering complaint details from citizens, processing the submitted information using AI-assisted techniques, categorizing issues into relevant departments, analyzing complaint similarities, and mapping reported issues based on their geographical locations. A centralized administrative dashboard enables authorities to review complaints, update their status, assign appropriate departments, and monitor resolution progress. The platform is developed using the MERN stack, with systematic testing and validation to evaluate its functionality, reliability, and effectiveness in supporting transparent and responsive civic grievance management.

3.2 General Objective

To develop a centralized crowdsourced civic complaint management platform that enables citizens to report civic issues and helps municipal departments efficiently classify, assign, monitor and resolve them.

3.3 Specific Objectives

The main objective of the CivicEye system is to enable citizens to digitally report civic issues by providing relevant details, location information, and supporting evidence.

The system aims to detect duplicate complaints using semantic text similarity and automatically classify complaints into appropriate civic categories. It also determines the severity and priority of complaints to help ensure that critical issues receive timely attention. Based on the complaint category, the system routes each complaint to the appropriate department, where the Department Head can assign it to a verifier for inspection and further action. The system provides controlled, role-based access to ensure that users can access only the information and functionalities permitted by

their roles. It also enables citizens to track complaint progress and confirm resolution, promoting transparency and accountability. Additionally, CivicEye provides centralized administrative monitoring and analytics to help administrators oversee departmental activities, monitor complaint status, and evaluate overall system performance.

3.4 Methodology

The project was carried out in phases that are visible in the individual contributions of the team: (i) study of existing platforms and civic portals, (ii) finalization of requirements, user workflow, database schema and REST API endpoints, (iii) development of the citizen frontend, the administrative dashboards and the backend services, (iv) revision of the design after supervisor feedback, and (v) demonstration and testing of each module. Supervisor feedback led to a simplified complaint submission flow, a revised dashboard design and refinements to the database and complaint lifecycle.

The operational methodology of the system is the complaint workflow shown in Figure 3.1. A citizen submits a complaint; the system validates it and detects duplicates; the complaint is assigned to a department; its case status is updated as it is handled; and it is finally resolved or closed. The overall CivicEye chain of responsibility is Citizen → Department Head → Verifier → Resolution → Citizen Confirmation.

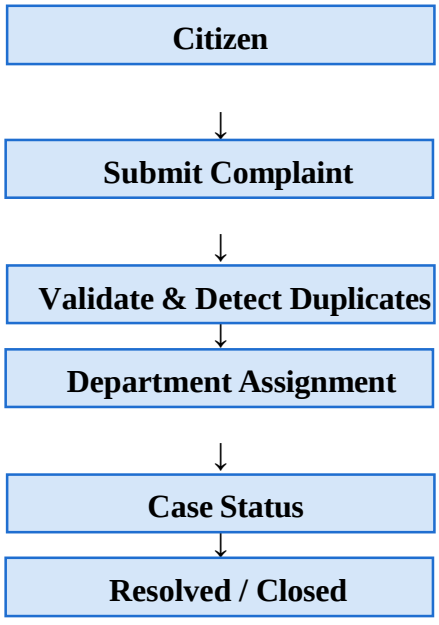


Figure 3.1: CivicEye Complaint Workflow

The stage “Citizen Confirmation” of the main workflow, in which the citizen confirms that the issue has been resolved.

3.5 Requirement Analysis

Table 3.1: User Roles, Responsibilities and Permissions

Role	Responsibilities	Permissions / access
Citizen	Registers and logs in; submits complaints with location and evidence; views and tracks own complaints; confirms resolution.	Own complaints only. Dashboard items visible in the prototype: Overview, Report Issue, Track Status, Identity Node, Help Center.
Department Head	Receives department-specific complaints; reviews details; assigns complaints to verifiers; updates official status; views department analytics.	Complaints of own department; may access citizen contact information when required. Dashboard items: Dept Dashboard, Department Queue, Civic Analytics, Help Center.
Verifier	Views complaints assigned by the Department Head; inspects issues in the field; reports status to the Department Head.	Only complaints assigned to the verifier. Workflow to be completed.
Super Admin	Monitors all departments; views overall progress; accesses department-wise analytics; inspects system performance.	System-wide monitoring; city-wide complaint management, analytics, priority reassignment and AI classification review (architecture view).

3.5.1 Functional Requirements

The functional requirements derived from the scope and objectives are listed in Table 3.1 together with their current status.

3.5.2 Non-functional Requirements

The system focuses on usability, security, scalability, and maintainability. Usability is ensured through a simple and user-friendly interface that allows citizens to submit complaints with minimal training. Security is maintained using JWT-based authentication, bcrypt password hashing, OTP verification for citizens, role-based

access control, and departmental scope isolation to protect user information and prevent unauthorized access. Scalability is supported through cloud deployment, allowing the system resources to be increased based on usage and demand.

Maintainability is achieved through modular services separated according to their responsibilities within a three-tier architecture, using open-source technologies.

Table 3.1 describes the roles, responsibilities, and permissions of users in the CivicEye system. Citizens can submit and track complaints, while Department Heads manage complaints and assign them to verifiers. Verifiers inspect assigned complaints and report their status. The Super Admin monitors all departments and views overall progress and analytics. These roles help ensure proper complaint management, secure access, and efficient coordination between citizens and government departments.

Table 3.2: Three-Tier Architecture of CivicEye

Layer	Components
Presentation layer	Citizen Dashboard, Department Dashboard, Verifier Interface, Super Admin Dashboard.
Application layer	Authentication, Complaint Management, Duplicate Detection, Classification, Priority Engine, Department Assignment, Analytics, Notifications.
Data layer	MongoDB: User Data, Complaint Data, Department Data, Classification Data, Status History.

3.5.3 User Requirements

The four roles and their responsibilities are summarized in Table 3.2.

3.5.4 System Constraints

The initial stage focuses on developing a working prototype of the proposed system. However, certain features, including OTP authentication, privacy controls, the verifier workflow, access restrictions, resolution confirmation, and notifications, are yet to be completed. The accuracy of duplicate complaint detection and complaint classification depends on external AI services and has not yet been evaluated.

Furthermore, an evaluation dataset has not been documented at this stage and will be added in the subsequent development phase. As specified in the interface submission

guide , image uploads in the current prototype are restricted to JPG, PNG, and WEBP formats, with a maximum file size of 5 MB.

3.6 Feasibility Analysis

3.6.1 Technical Feasibility

The proposed system uses a web-based architecture with a React-based frontend to provide a user-friendly interface for citizens and administrative users. The backend is developed using Node.js and Express.js, which handle business logic and provide REST APIs for communication between the frontend and backend. MongoDB is used as the database to store complaint details, user information, and related records. JWT-based authentication is implemented to ensure secure user access and authentication. Semantic text embeddings are used for duplicate complaint detection by identifying similarities in the meaning of complaint descriptions, helping the system recognize and group related complaints efficiently. Additionally, AI-assisted classification with fallback logic is incorporated to automatically categorize complaints based on their content. If the AI service fails or is unavailable, the fallback mechanism helps ensure that complaints can still be processed and assigned an appropriate category.

3.6.2 Operational Feasibility

The proposed system provides a simple and user-friendly process for citizens to submit complaints with minimal effort. It supports department-specific workflows to ensure that complaints are directed to the appropriate departments for further action. Role-based dashboards are provided for citizens, administrators, department heads, and staff members, allowing each user to access features according to their responsibilities. The system also enables centralized complaint tracking, allowing users and administrators to monitor complaint status and progress efficiently. Overall, the interface is designed to be easy to understand and operate, requiring minimal training for citizens and administrative users.

3.6.3 Economic Feasibility

The proposed system uses open-source technologies, making development and maintenance more cost-effective. Cloud deployment allows the system to scale its resources based on usage and demand, ensuring flexibility as the number of users and complaints increases. No specialized hardware is required, as the system can operate

using standard computing devices with internet access. Additionally, the automated complaint classification, department-specific workflows, and centralized complaint tracking help reduce manual administrative effort, improve operational efficiency, and simplify complaint management.

3.7 System Architecture

CivicEye follows a three-tier architecture, summarized in Table 3.3, and illustrated by the system architecture diagram in Figure 3.2.

Table 3.2 describes the three-tier architecture of the CivicEye system. The presentation layer provides dashboards and interfaces for citizens, Department Heads, verifiers, and the Super Admin. The application layer handles authentication, complaint management, duplicate detection, classification, department assignment, and notifications. The data layer uses MongoDB to store user details, complaints, department information, and complaint status history. These three layers work together to ensure efficient complaint management and smooth system operation.

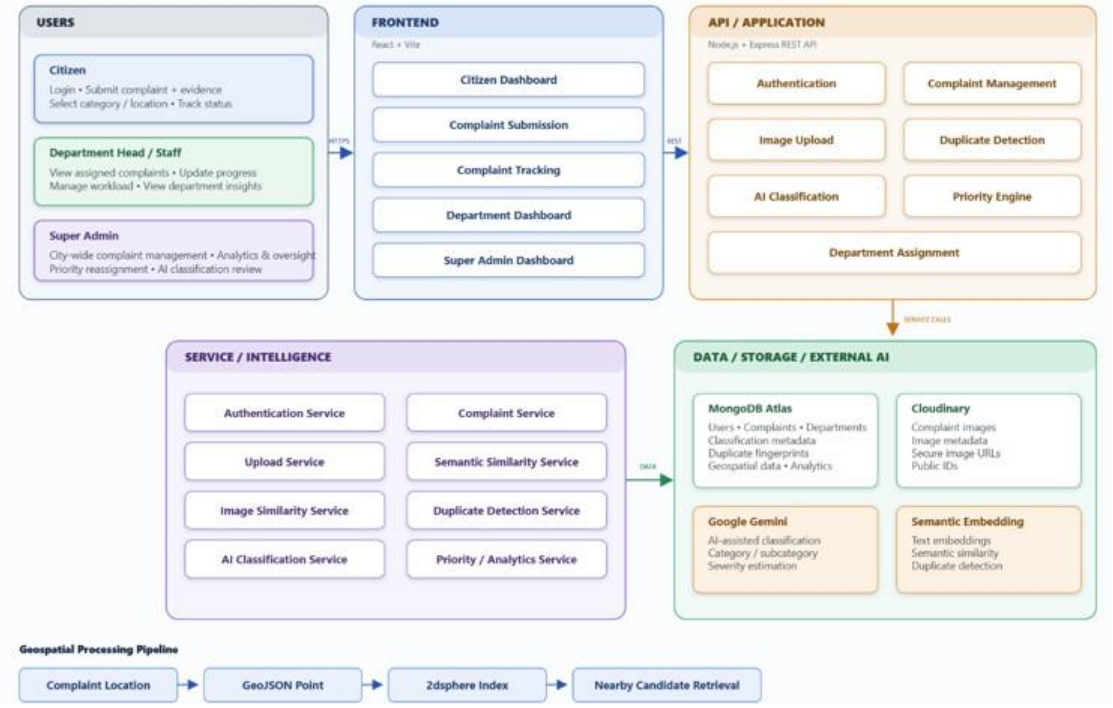


Figure 3.2: CivicEye System Architecture

In the architecture diagram, users (Citizen, Department Head/Staff and Super Admin) reach the React and Vite frontend over HTTPS. The frontend exposes the Citizen Dashboard, Complaint Submission, Complaint Tracking, Department Dashboard and Super Admin Dashboard, and communicates over REST with the Node.js and Express API layer. The API layer contains Authentication, Complaint Management, Image Upload, Duplicate Detection, AI Classification, Priority Engine and Department Assignment. Service calls reach a service and intelligence layer (Authentication, Complaint, Upload, Semantic Similarity, Image Similarity, Duplicate Detection, AI Classification, and Priority/Analytics services), which uses the data, storage and external AI layer: MongoDB Atlas (users, complaints, departments, classification metadata, duplicate fingerprints, geospatial data, analytics), Cloudinary (complaint images, image metadata, secure image URLs, public IDs), Google Gemini (AI-assisted classification, category and subcategory, severity estimation) and semantic embedding (text embeddings, semantic similarity, duplicate detection).

3.8 Detailed Module Description

The core modules are Citizen Management, Complaint Management, Authentication and Authorization, Complaint Categorization, Duplicate Detection, Department Management, Verifier Management, Priority Management, Status Management, Notification and Alert Management, Analytics and Administrative Monitoring. Each is described in Chapter IV.

3.9 Algorithms and Techniques

Table 3.3: Algorithms and Techniques

Function	Technique
Duplicate detection	Semantic embeddings and similarity comparison; supported by geospatial proximity and image fingerprinting.
Complaint classification	AI-assisted classification with a centralized civic taxonomy; rule-based fallback classification.
Priority calculation	Deterministic calculation from severity, urgency, safety risk and public impact.
Access control	Role-Based Access Control (RBAC) with departmental scope isolation.
Complaint routing	Category → Department mapping.
Authentication	JWT-based session authentication with OTP verification for citizens.

Table 3.3 presents the algorithms and techniques used in the CivicEye system. Duplicate detection uses semantic similarity, location, and image comparison to identify repeated complaints. AI-assisted classification helps categorize complaints, while priority calculation considers severity, urgency, safety risk, and public impact. Role-Based Access Control ensures secure access, and category-based mapping assigns complaints to the appropriate departments. JWT authentication is used for user login, while OTP verification is planned for citizens.

3.9.1 Duplicate Detection

Duplicate Detection identifies multiple complaints submitted by different citizens about the same civic issue. The system compares complaint descriptions, categories, and geographical locations to detect similar reports and flag potential duplicates. For example, several citizens reporting the same pothole from nearby locations can be identified as related complaints. This helps CivicEye reduce redundant reports, consolidate information about recurring issues, avoid unnecessary workload for municipal authorities, and improve the efficiency of complaint tracking and resolution.

3.9.2 Complaint Classification

The classification flow is: Complaint → AI-assisted classification → Civic category → Department mapping. The AI service (Google Gemini) proposes a category and subcategory from a centralized civic taxonomy. If the AI service cannot be used, a rule-based fallback classification is applied so that the complaint can still be routed. Administrators have an AI classification review and manual correction workflow. The mapping from category to department decides which Department Head receives the complaint. Taxonomy values visible in the prototype include Roads & Infrastructure (subcategory Pothole), Streetlights & Electricity, Waste Management & Sanitation, Water Supply and Public Safety & Other Civic Hazards.

3.9.3 Severity and Priority

Severity is predicted by the AI-assisted component. Priority is then calculated deterministically from four factors: **severity, urgency, safety risk and public impact**.

3.9.4 Role-Based Access Control and Authentication

Access is controlled by RBAC with departmental scope isolation, so that a Department Head sees only the complaints of that department. Sessions are authenticated with JSON Web Tokens, passwords are protected using bcrypt, and OTP verification is planned for citizens.

3.10 Data Flow Architecture

The complaint submission process begins when the citizen enters the complaint title, category, subcategory, and description, selects the location on the map using GPS or search, and optionally uploads an image. The React-based frontend sends the complaint details to the REST API using Axios, with the user session identified through JWT authentication. The API validates the request and stores the uploaded evidence image through the Cloudinary-based upload service. The geospatial pipeline stores the complaint location as a GeoJSON Point and retrieves nearby complaints using a 2dsphere index. The duplicate detection service compares the new complaint with existing nearby complaints using semantic similarity and image fingerprints to identify potential duplicates. Next, the AI classification service assigns a category and subcategory and estimates the severity of the complaint. The priority engine then determines the complaint priority based on severity, urgency, safety risk, and public impact. The complaint is routed to the appropriate department using the category-to-department mapping and appears in that department's queue. The Department Head assigns a verifier to inspect the complaint, while status updates are recorded in the status history, allowing citizens to track progress. Once the complaint is resolved, the citizen is expected to confirm the resolution.

The geospatial portion of this flow is shown in Figure 3.3.

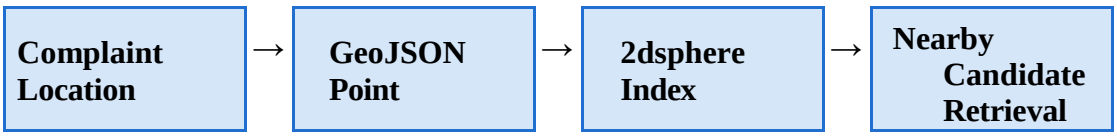


Figure 3.3: Geospatial Processing Pipeline

3.11 System Integration

The React frontend (Vite, Tailwind CSS) is integrated with the backend through Axios calls to REST endpoints, with React Context managing application state. The backend integrates MongoDB Atlas through Mongoose, Cloudinary for image

storage, Google Gemini for AI-assisted classification and severity estimation, and a semantic embedding service for duplicate detection.

3.12 Testing and Validation

Testing and Validation ensure that CivicEye functions correctly, reliably, and efficiently under different conditions. The system is tested to verify citizen registration and login, complaint submission, AI-assisted complaint classification, duplicate detection, geolocation accuracy, and administrative complaint management. Functional testing checks whether each feature behaves as expected, while integration testing verifies communication between the frontend, backend, and database. Validation also involves checking the accuracy of complaint categorization, the identification of potential duplicate reports, and the correctness of complaint status updates. These tests help identify errors, improve system reliability, and ensure that CivicEye provides a user-friendly and effective platform for civic issue reporting and resolution.

3.13 Performance Considerations

The use of a 2dsphere index restricts duplicate comparison to nearby candidate complaints rather than all complaints, and the cloud deployment can be scaled according to usage. Measured response times, throughput and duplicate-detection latency have not been recorded.

3.14 Error Handling and Security

The system implements JWT-based authentication and bcrypt password hashing to provide secure user authentication and session management. Authorization is managed through role-based access control and departmental scope isolation, ensuring that users can access only the features and information permitted by their roles and departments. For media uploads, the interface accepts evidence images in JPG, PNG, or WEBP formats with a maximum file size of 5 MB, and the images are stored using Cloudinary. The system also incorporates a rule-based fallback mechanism to classify complaints when AI-assisted classification is unavailable, ensuring that complaint

processing can continue. Citizen phone-number privacy controls are yet to be finalized to strengthen the protection of personal information.

3.15 Challenges and Solutions

Table 3.4: Challenges and the CivicEye Response

Challenge	CivicEye response
Manual or fragmented reporting	Single web platform for submission with location and evidence.
Repeated complaints for the same issue	Semantic duplicate detection with geospatial proximity and image fingerprinting.
Incorrect or inconsistent categorization	AI-assisted classification against a centralized civic taxonomy, rule-based fallback and manual correction workflow.
Difficulty prioritizing critical issues	Severity prediction and deterministic priority calculation.
Delayed assignment	Category → department routing and Department Head assignment to verifiers.
Limited transparency for citizens	Complaint tracking with lifecycle progress.
Lack of centralized monitoring	Department dashboards, Super Admin monitoring and analytics.

Table 3.4 explains the major challenges in civic complaint management and how CivicEye addresses them. The system provides a single platform for reporting issues with location and evidence. It helps detect duplicate complaints, classify issues, prioritize critical problems, and assign complaints to the concerned departments. Citizens can track complaint progress, while department dashboards and Super Admin monitoring improve transparency and overall management.

3.16 Future Enhancements

The planned future work focuses on completing the remaining features and improving the overall system. Further improvements will include enhancing notification mechanisms, completing system integration, conducting comprehensive security testing, and optimizing deployment. In addition, experimental evaluation will be carried out to measure duplicate detection performance, complaint classification accuracy, and overall system efficiency.

CHAPTER IV

PROPOSED WORK/MODULES

4.1 Introduction

CivicEye proposes an AI-assisted civic grievance management platform that enables citizens to report public issues and helps local authorities manage and resolve them efficiently. The system integrates complaint registration, AI-assisted complaint classification, duplicate detection, geospatial issue mapping, urgency-based prioritization, and centralized administrative management into a unified platform. Citizens can submit complaints related to road damage, water supply, waste management, and sanitation, along with relevant descriptions, images, and location details. The system processes these reports to categorize issues, identify potential duplicate complaints, and support authorities in assigning and tracking cases until resolution. The proposed modules aim to reduce manual effort, improve transparency, minimize redundant reports, and enable data-driven decision-making for effective civic issue management.

4.2 Citizen Management

This module covers secure registration and authentication of citizens and the views a citizen needs. The citizen navigation visible in the prototype consists of Overview, Report Issue, Track Status, Identity Node and Help Center. A citizen can view only personally submitted complaints. Registration, login, reporting and tracking were tested by the team member responsible for the citizen module.

4.3 Complaint Management

The complaint management module accepts a complaint with a title, main category, subcategory, narrative, location and optional image evidence, stores it, and maintains it through its lifecycle. The citizen-facing lifecycle shown in the tracking interface has the stages Submitted, Assigned, In Progress and Resolved. Each complaint receives a tracking identifier of the form #CMP followed by digits, which is used in both the citizen tracking view and the department queue. On the department side, complaints can be filtered by status, SLA status, escalation and priority, and searched by identifier, title or department.

4.4 Authentication and Authorization

Authentication uses JWT-based session tokens, crypt for password hashing and, for citizens, OTP verification. Authorization uses role-based access control for the four roles Citizen, Department Head, Verifier and Super Admin, with departmental scope isolation so that department users access only their department's complaints. The role is visible in the interface, where the logged-in user card shows "Citizen" or "Department head". Department-wise access restrictions are to be finalized.

4.5 Complaint Categorization

Each complaint is assigned a main category and a subcategory from a centralized civic taxonomy. The report form shows Main Category and Subcategory fields. AI-assisted classification using Google Gemini proposes the category and subcategory; when the AI service is unavailable a rule-based fallback classification is used. An AI classification review and manual correction workflow allows administrators to correct a classification, and department configuration and routing rules map the category to the department.

4.6 Duplicate Detection

The duplicate detection module combines semantic text similarity, geospatial proximity and image fingerprinting. Sentence embeddings are compared using cosine similarity; nearby candidates are retrieved using the 2dsphere index; and image fingerprints are compared for uploaded evidence. The citizen interface presents a Duplicate Warning Modal when a probable duplicate is detected. The interface also advertises "Intelligent Duplicate Detection" in the Citizen Reporting Intelligence panel and lists "Nearby Complaints (500m Radius)" while the citizen is choosing a location.

4.7 Department Management

The department module supports department configuration and routing rules, so that complaints reach the appropriate department by category-to-department mapping. Each Department Head works from a department-specific Command Center that contains a summary of KPIs, queue tabs for All Complaints, Unassigned Queue and Assigned Queue, workflow filters and an operations matrix listing complaint identifier, issue details, department, assigned verifier, priority, SLA state, status and actions.

4.8 Verifier Management

The Department Head assigns complaints to department verifiers using the Assign Verifier action, and may use Reassign Verifier once a complaint has been assigned (Figure 5.4). A verifier is expected to view complaints assigned to them, inspect the reported civic issue, perform field verification and report the issue status to the Department Head. The Head/Verifier workflows have been developed in the backend.

4.9 Priority Management

Priority is computed deterministically from severity, urgency, safety risk and public impact, where severity is predicted by the AI-assisted component. The resulting priority is displayed in the department views as a priority badge and counted in KPI cards. The Super Admin can reassign priority. No numeric formula is specified .

4.10 Status Management

Complaint status moves through the lifecycle and is recorded in the status history stored in the data layer. In the department matrix, status badges such as SUBMITTED and ASSIGNED are shown, and a separate SLA State badge (for example ON TRACK) is visible. The Department Head updates the official complaint status, and the citizen sees the current status in the tracking view.

4.11 Notification and Alert Management

A notification indicator is present in the header of every screen of the prototype (a bell icon with a count badge). The module provides notifications and alerts to keep citizens and administrators informed about important complaint updates and status changes. It helps improve communication and ensures better tracking of reported civic issues.

4.12 Analytics

The Civic Analytics page presents KPI cards, a complaint volume time-series trend with a daily interval and period-over-period change, a Top Main Categories bar chart, and a Department Performance and Workload Matrix. It is reachable from the Department Head Command Center through the “Civic Analytics & Intelligence” button (Figure 5.4).

4.13 Administrative Monitoring

The Super Admin monitors all departments, views overall system progress, accesses department-wise analytics and inspects system performance. The architecture view lists city-wide complaint management, analytics and oversight, priority reassignment and AI classification review for this role. Administrative dashboards, complaint monitoring, filtering and department-wise progress tracking are listed among the implemented contributions.

4.14 AI / Intelligent Components

Table 4.1: Intelligent Components

Component	Role in CivicEye
Semantic text embeddings	Represent complaint text as vectors so that complaints with similar meaning can be compared.
Similarity-based duplicate detection	Compares a new complaint with nearby existing complaints using embedding similarity, proximity and image fingerprints.
AI-assisted complaint classification	Google Gemini proposes category and subcategory from the centralized civic taxonomy.
Severity prediction	AI-assisted estimate of how severe the reported issue is.
Deterministic priority calculation	Combines severity, urgency, safety risk and public impact into a priority.
Rule-based fallback classification	Categorizes complaints when the AI service is unavailable.

Table 4.2 describes the intelligent components used in the CivicEye system. Semantic embeddings and similarity-based detection help identify duplicate complaints. AI-assisted classification categorizes complaints, while severity prediction helps identify serious issues. The priority calculation considers urgency, safety risk, and public impact. A rule-based fallback is also used when the AI service is unavailable. These components help improve complaint classification and management.

4.15 Geospatial Processing

The citizen selects the complaint location on a Leaflet map (OpenStreetMap tiles) by dragging a marker, using the browser's geolocation ("Use My Current Location", which also reports a GPS fix accuracy) or searching for a street, landmark, city or pincode.

The system performs reverse geocoding to produce a formatted location address with latitude, longitude, locality and city/state (Figure 5.2). On the server the location is stored as a GeoJSON Point with a 2dsphere index so that nearby candidates can be retrieved.

4.16 System Integration

The modules are integrated through REST APIs between the React frontend and the Node.js/Express backend, with MongoDB Atlas as the shared data store, Cloudinary for evidence images, Google Gemini for classification and severity estimation and a semantic embedding service for duplicate detection.

4.17 Summary

CivicEye integrates multiple modules to simplify civic issue reporting and improve grievance management. The platform enables citizens to submit complaints, while AI-assisted classification, duplicate detection, and geospatial issue mapping help organize and analyze reported problems. The administrative dashboard supports complaint assignment, prioritization, status tracking, and resolution monitoring. Additionally, notification and alert management helps keep users informed about important updates. Together, these modules aim to reduce manual effort, improve transparency, and support faster, more efficient civic issue resolution.

CHAPTER V

RESULTS AND DISCUSSION

5.1 Citizen Module Results

The citizen portal uses a left navigation with Overview, Report Issue, Track Status, Identity Node and Help Center items, a universal search box, a notification indicator and the current date in the header, and a user card showing the logged-in user ("Demo Citizen", role "Citizen") with a Terminate Session action (Figures 5.1 and 5.3). The Report Issue and Track Status screens, which are the main citizen functions, are presented in the next two sections.

5.2 Complaint Submission Results

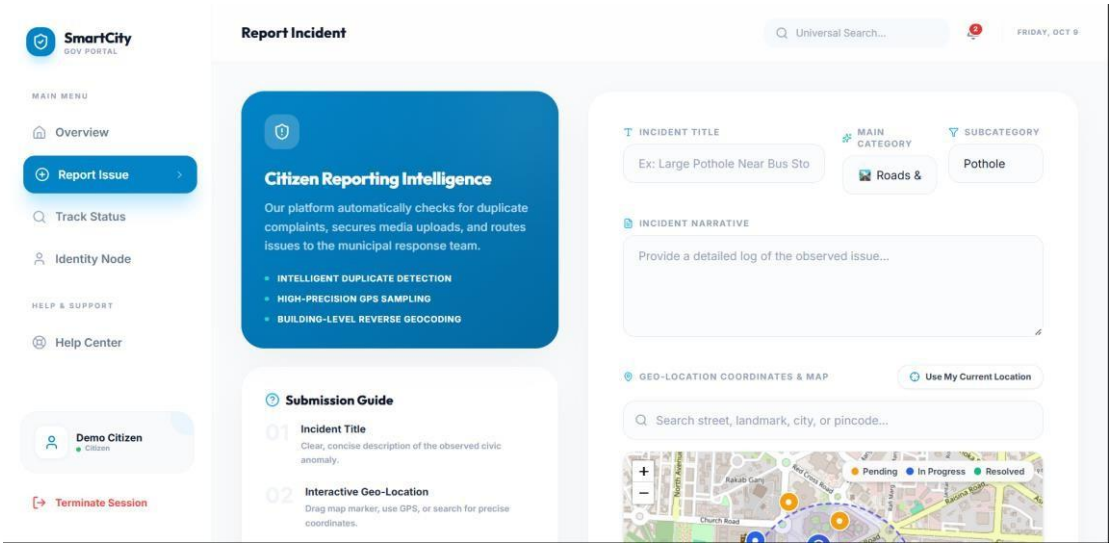


Figure 5.1: Citizen Report Incident Interface

Figure 5.1 shows the Report Incident screen. The left-hand "Citizen Reporting Intelligence" panel states that the platform automatically checks for duplicate complaints, secures media uploads and routes issues to the municipal response team, and lists Intelligent Duplicate Detection, High-Precision GPS Sampling and Building-Level Reverse Geocoding. The form contains the Incident Title, Main Category, Subcategory and Incident Narrative fields, followed by the geo-location section. In this screenshot Main Category and Subcategory show a Roads & Infrastructure category and the subcategory Pothole.

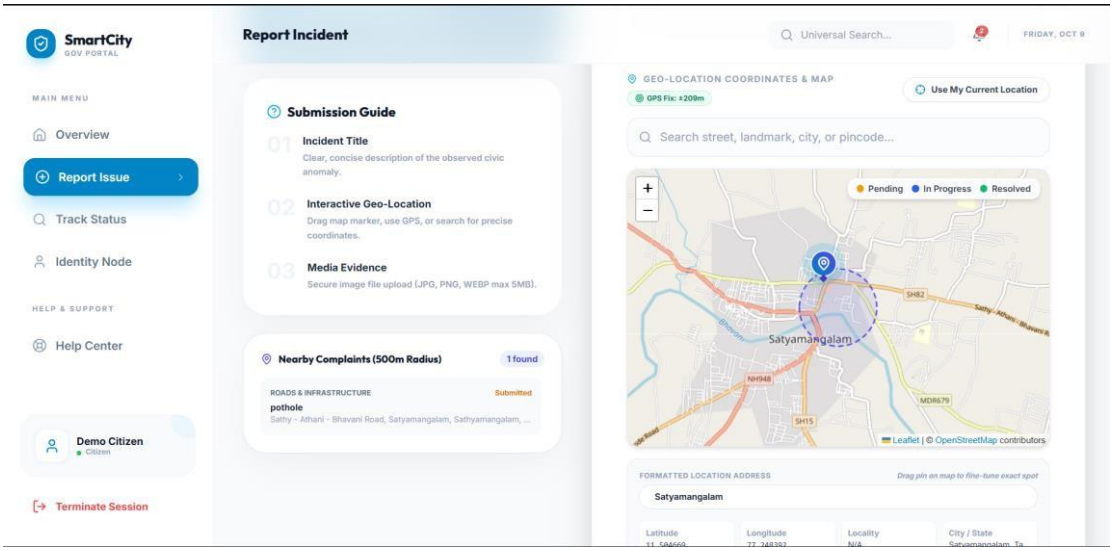


Figure 5.2: Geo-Location Capture with Interactive Map, GPS Fix and Nearby Complaints (500 m Radius)

Figure 5.2 shows the geo-location section and the Submission Guide. The guide explains the three submission steps: Incident Title, Interactive Geo-Location (drag the map marker, use GPS, or search for precise coordinates) and Media Evidence (secure image upload, JPG, PNG or WEBP, maximum 5 MB). In Progress and Resolved complaints. A GPS Fix accuracy indicator and a "Use My Current Location" button are provided, and the Formatted Location Address area shows the reverse-geocoded address (here "Satyamangalam") together with latitude and longitude.

5.3 Complaint Tracking Results

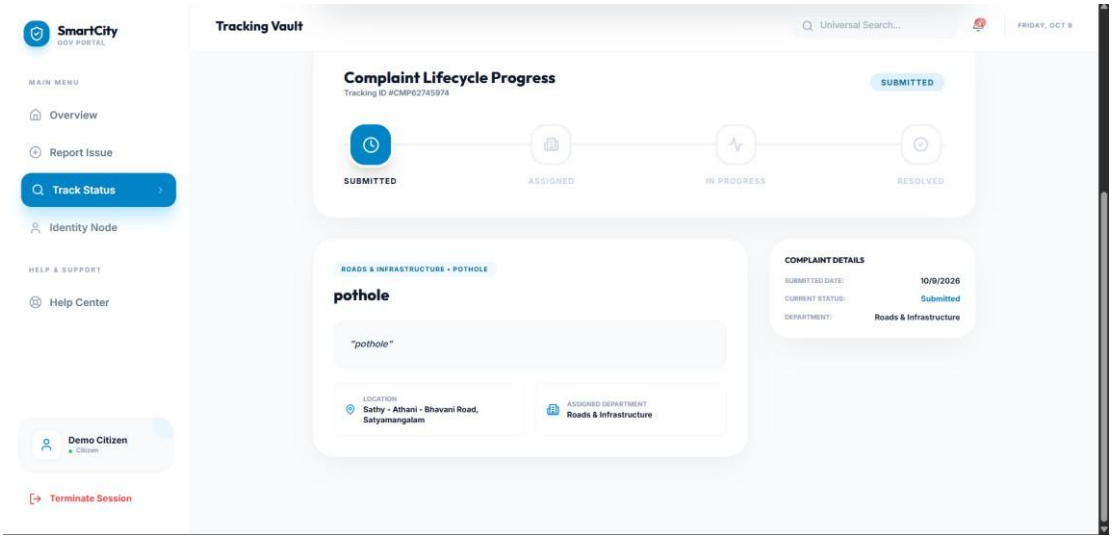


Figure 5.3: Citizen Complaint Tracking Interface (Tracking Vault)

Figure 5.3 shows the Tracking Vault. The Complaint Lifecycle Progress component displays the tracking identifier (#CMP62745974) and four stages: Submitted, Assigned, In Progress and Resolved, with the current stage (Submitted) highlighted. Below it, the complaint details show the category label "Roads & Infrastructure • Pothole", the title and narrative, the location (Sathy – Athani – Bhavani Road, Satyamangalam), the assigned department (Roads & Infrastructure), the submitted date, and the current status.

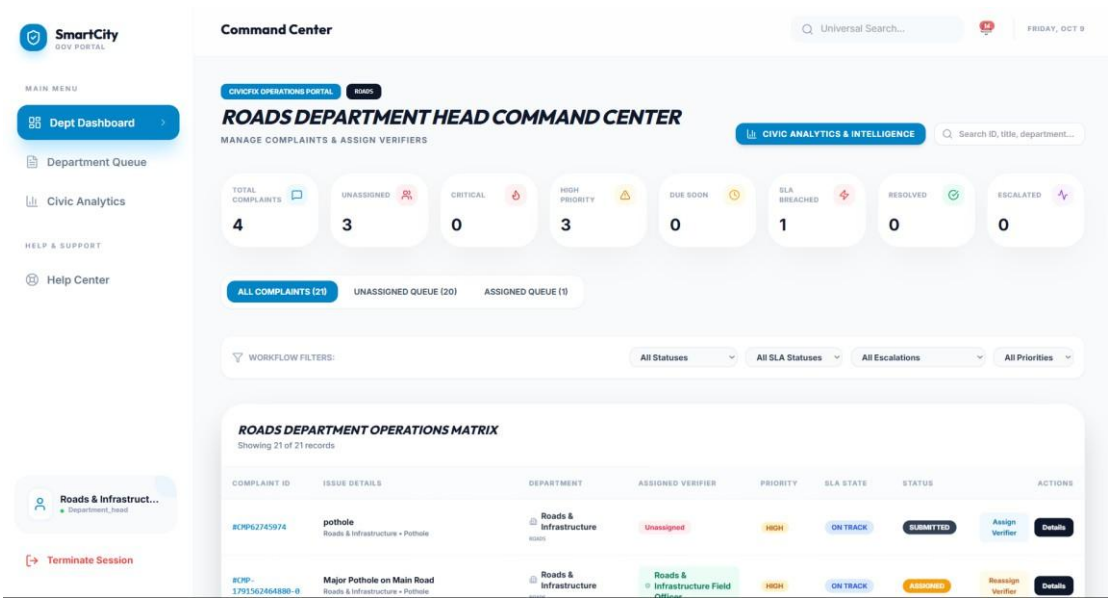


Figure 5.4: Department Head Command Center with Operations Matrix and Verifier Assignment

The same complaint (#CMP62745974) appears in the Roads department queue in Figure 5.4 with the status SUBMITTED, which shows that the complaint submitted by the citizen reaches the department that was assigned to it.

5.4 Duplicate Detection Interface

The duplicate detection engine and the Duplicate Warning Modal are part of the implemented contributions. In the screenshots available, duplicate awareness is visible in two places: the Citizen Reporting Intelligence panel that lists Intelligent Duplicate Detection (Figure 5.1) and the "Nearby Complaints (500m Radius)" panel, which showed one existing complaint (Roads & Infrastructure, "pothole", Submitted) while the citizen was selecting a location (Figure 5.2).

5.5 Department Head Interface

Figure 5.4 shows the Roads Department Head Command Center ("Manage complaints & assign verifiers"). The page contains a row of KPI cards (Total Complaints, Unassigned, Critical, High Priority, Due Soon, SLA Breached, Resolved, Escalated), a "Civic Analytics & Intelligence" button, a search box, queue tabs (All Complaints, Unassigned Queue, Assigned Queue), workflow filters (All Statuses, All SLA Statuses, All Escalations, All Priorities) and the Roads Department Operations Matrix. The matrix lists the complaint identifier, issue details, department, assigned verifier, priority, SLA state, status and actions. The first row is the citizen complaint "pothole" (#CMP62745974), shown as Unassigned with HIGH priority, ON TRACK SLA and SUBMITTED status, with the actions Assign Verifier and Details. The second row, "Major Pothole on Main Road", is assigned to a Roads & Infrastructure field officer, with the actions Reassign Verifier and Details.\

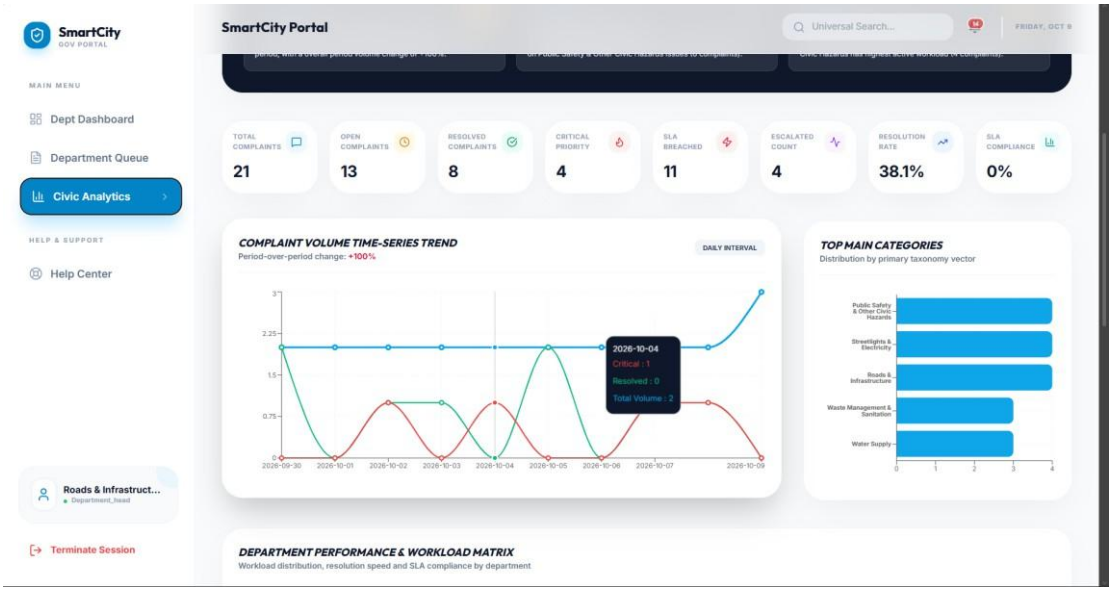


Figure 5.5: Civic Analytics Dashboard – KPIs, Complaint Volume Trend and Top Main Categories

Figure 5.5 shows the Civic Analytics page for the same department head. The KPI cards show Total Complaints, Open Complaints, Resolved Complaints, Critical Priority, SLA Breached, Escalated Count, Resolution Rate and SLA Compliance. The Complaint Volume Time-Series Trend chart plots daily Critical, Resolved and Total Volume series, with a tooltip for a selected day and a period-over-period change indicator. The Top Main Categories chart shows the distribution by primary taxonomy category

(Public Safety & Other Civic Hazards, Streetlights & Electricity, Roads & Infrastructure, Waste Management & Sanitation and Water Supply), and the Department Performance and Workload Matrix section follows

The Department Head side of verifier management is visible in Figure 5.4 (Assigned Verifier column, Assign Verifier and Reassign Verifier actions). The verifier's own interface, in which assigned complaints are viewed, inspected and reported back.

5.6 Super Admin Interface

The Super Admin role monitors all departments, overall progress, department-wise analytics and system performance. The analytics views in Figures 5.4 and 5.5 are those of a Department Head.

5.7 AI-assisted Classification

AI-assisted classification, severity prediction and priority calculation are implemented as backend services, with a manual correction workflow for administrators and a rule-based fallback. The form in Figure 5.1 provides the category and subcategory fields, and the department views present the resulting category (for example "Roads & Infrastructure • Pothole") and priority badge (Figure 5.4).

5.8 Geospatial Features

The geospatial features visible in the prototype are the Leaflet map with a draggable marker, GPS fix indicator, address search, reverse geocoding, latitude and longitude display, a search radius circle and nearby complaint listing (Figure 5.2). Complaint locations are stored as GeoJSON points with a 2dsphere index for nearby candidate retrieval (Figure 3.3).

5.9 System Workflow

Taken together, the screenshots demonstrate the following part of the main workflow: a citizen reports an issue (Figures 5.1 and 5.2), the complaint receives a tracking identifier and the status Submitted (Figure 5.3), it appears in the responsible department's queue and can be assigned to a verifier (Figure 5.4), and the department can analyse its complaints (Figure 5.5).

5.10 Testing and Validation

Testing and validation are performed to ensure that CivicEye modules function correctly, including complaint submission, classification, duplicate detection, and status tracking. The system is evaluated for functionality, reliability, and accuracy to ensure effective civic grievance management.

5.11 Discussion

This show that the core citizen-to-department path is working in the prototype: location-aware reporting, tracking, department routing and department-side management with analytics. The design places intelligence (duplicate detection, classification, priority) behind a simple form, which is consistent with the operational-feasibility goal of minimal training. Because the platform's intelligent behaviour has not yet been measured, claims about reduced duplicates, faster routing or better prioritization remain expectations rather than results.

5.12 Current Limitations

CivicEye currently has limitations in the accuracy of its AI-assisted complaint categorization, as it relies on keyword-based analysis that may misclassify complex or ambiguously worded complaints. The platform depends on citizens providing accurate descriptions, images, and location details, which may affect complaint verification and prioritization. Real-time integration with municipal systems is limited, and complaint assignment, status updates, and resolution depend on the availability and active participation of authorities. The system also requires further improvements in scalability, multilingual support, duplicate complaint detection, data privacy, and automated urgency assessment to ensure reliable performance in real-world environments.

CHAPTER VI

CONCLUSION AND FUTURE WORK

6.1 Conclusion

CivicEye provides an integrated and user-friendly platform for efficient civic issue reporting and grievance management by connecting citizens with local authorities. The system incorporates AI-assisted complaint classification, duplicate detection, geospatial issue mapping, and centralized administrative management to simplify the handling of public complaints. By organizing reported issues, identifying similar complaints, and enabling authorities to monitor complaint status and resolution progress, CivicEye aims to reduce manual effort and improve transparency in public service delivery. The proposed system also supports data-driven decision-making and better identification of recurring civic problems, contributing to more effective resource allocation and improved community well-being. With further enhancements and real-world implementation, CivicEye has the potential to strengthen communication between citizens and authorities and promote a more responsive, accountable, and efficient civic administration system.

6.2 Achievements

The CivicEye system is developed using a three-tier architecture consisting of a React/Vite frontend, a Node.js/Express REST API backend, and a MongoDB Atlas data layer. The Citizen module provides features such as registration, login, complaint reporting, complaint tracking, and complaint history viewing. An interactive Leaflet map with geolocation and reverse geocoding enables citizens to accurately capture the location of civic issues. The system incorporates a semantic duplicate detection engine that uses sentence embeddings, cosine similarity, geospatial proximity, and image fingerprinting to identify potentially repeated complaints, with a Duplicate Warning Modal displayed on the frontend. AI-assisted classification is used to categorize complaints, predict severity, calculate priority, and route complaints to the appropriate departments. Backend security is implemented through JWT-based authentication, role-based access control (RBAC), and departmental scope isolation to ensure controlled access to complaint information. Additionally, administrative

dashboards provide complaint monitoring, filtering, department-wise progress tracking, and analytics. The system also supports an AI classification review and manual correction workflow, allowing administrators to review and correct automatically assigned complaint categories when necessary.

6.3 Future Scope

In the future, CivicEye can be enhanced by incorporating advanced natural language processing techniques to improve complaint classification and semantic duplicate detection, enabling the system to identify similar complaints even when they are expressed differently. Geospatial analysis can be further developed to identify civic issue hotspots, detect recurring infrastructure problems, and support data-driven maintenance planning. Integration with municipal grievance management systems could enable automated complaint assignment, real-time status synchronization, and more effective coordination between citizens and government departments. Additional features such as multilingual complaint submission, image-based issue verification, automated notifications, and citizen feedback mechanisms could further improve accessibility and transparency. Furthermore, strengthening data security, privacy protection, and system scalability would support reliable operation across larger geographical areas and accommodate increasing numbers of users and complaints. These enhancements could help CivicEye evolve into a comprehensive civic grievance management platform that promotes accountable governance and efficient public service delivery.

APPENDICES

I. SAMPLE CODE

```
const fetchData = async () => {
    setLoading(true);
    try {
      const [metricsRes, complaintsRes, deptsRes] = await Promise.all([
        api.get('/admin/metrics'),
        api.get('/admin/complaints'),
        api.get('/departments')
      ]);
      setMetrics(metricsRes.data);
      setComplaints(complaintsRes.data);
      setDepartments(deptsRes.data);

      if (isDeptHead) {
        try {
          const staffRes = await api.get('/department/staff');
          setStaffMembers(staffRes.data.staffMembers || []);
        } catch (e) {
          console.error('Failed to fetch department staff list:', e);
        }
      }
    } catch (error) {
      console.error('Failed to load dashboard metrics:', error);
    } finally {
```

```
            setLoading(false);
        }
    };

    useEffect(() => {
        fetchData();
    },
[]);
```

APPENDIX A – TEST CASES

The test cases below are the planned scenarios derived from the scope of the project. The result column is intentionally left open; no test results are claimed.

Table A.1: Test Case Outline

ID	Test scenario	Expected behaviour	Result
TC1	Citizen registration and login	Account created; JWT session issued	Account created; JWT session issued
TC2	OTP verification for citizen (after implementation)	Only correct OTP completes verification	Only correct OTP completes verification
TC3	Submit complaint with location and image	Complaint stored; tracking ID shown	Complaint stored; tracking ID shown
TC4	Submit a near-identical complaint at the same place	Duplicate warning is presented	Duplicate warning is presented
TC5	Complaint classification with AI service unavailable	Rule-based fallback assigns a category	Rule-based fallback assigns a category
TC6	Priority calculation for different severities	Priority reflects severity, urgency, safety risk and public impact	Priority reflects severity, urgency, safety risk and public impact
TC7	Routing of a Roads complaint	Appears in the Roads department queue	Appears in the Roads department queue
TC8	Department Head assigns and reassigns a verifier	Assigned verifier shown in matrix	Assigned verifier shown in matrix
TC9	Department filters by status, SLA, escalation and priority	Matrix shows only matching records	Matrix shows only matching records
TC10	Department user opens another department’s complaint	Access denied (departmental scope isolation)	Access denied (departmental scope isolation)
TC11	Citizen tracks own complaint	Lifecycle stage and details shown	Lifecycle stage and details shown
TC12	Citizen confirms resolution (after implementation)	Complaint closed	Complaint closed

WORK CONTRIBUTION

The CivicEye project was a collaborative effort among three team members, with each member responsible for a specific module and contributing to the overall development, testing, and documentation of the system.

MEMBER 1:ARUNRAJAN N M (7376232CB103)-ADMIN MODULE

Weekly Log:

- Week 1: Studied administrative workflows in existing civic complaint systems, identified dashboard requirements, and prepared initial project documentation.
- Week 2: Reviewed existing administrator dashboards and documented complaint monitoring, verification, and status management requirements.
- Week 3: Finalized administrative requirements for complaint verification, filtering, status updates, and dashboard reporting.
- Week 4: Planned the administrator dashboard structure and documented the implementation workflow and system architecture.
- Week 5: Finalized dashboard features, including complaint filtering, status updates, verification, and report generation.
- Week 6: Revised the dashboard design based on supervisor feedback and improved complaint management and reporting workflows.
- Week 7: Compared administrative dashboard features with existing platforms and documented the comparative analysis.
- Week 8: Reviewed administrative workflows, verified dashboard documentation, and organized project design records.
- Week 9: Demonstrated the administrative dashboard, tested complaint management and status updates, and documented supervisor feedback.
- Week 10: Finalized dashboard features, tested complaint filtering and management functions, and completed project documentation and presentation materials.

MEMBER 2: RAJAGOPAL S (7376232CB139)-CITIZEN MODULE

Weekly Log:

- Week 1: Studied existing civic complaint portals and identified citizen requirements, including registration, complaint submission, and complaint tracking.
- Week 2: Analyzed existing user interfaces and documented functional requirements for simple complaint reporting and tracking.
- Week 3: Studied image-based and location-based complaint reporting and refined the citizen workflow.
- Week 4: Planned citizen portal navigation and designed basic layouts for registration and complaint submission pages.
- Week 5: Finalized the citizen module workflow and prepared the frontend implementation plan.
- Week 6: Reviewed the citizen interface based on supervisor feedback and identified improvements to complaint submission and navigation.a
- Week 7: Compared citizen-side features with existing platforms and refined the interface for better usability.
- Week 8: Reviewed citizen module documentation, verified the planned workflow, and identified minor interface improvements.
- Week 9: Demonstrated registration, login, complaint submission, and tracking features and verified frontend-backend integration.
- Week 10: Finalized the citizen module, tested registration and complaint reporting, improved the interface based on feedback, and verified frontend functionality.

MEMBER 3:FIRZAAN I (7376232CB117)-BACKEND DEVELOPER

Weekly Log:

- Week 1: Studied the complaint management workflow, identified major system entities and relationships, and analyzed backend requirements.
- Week 2: Reviewed existing complaint processing mechanisms, refined the problem statement, and planned the preliminary database structure and backend

workflow.

- Week 3: Finalized functional and non-functional requirements and identified validation, security, and database requirements.
- Week 4: Analyzed backend architecture and planned the database schema, REST API structure, and complaint lifecycle.
- Week 5: Finalized the database schema, planned API endpoints for authentication and complaint management, and documented the processing workflow.
- Week 6: Refined the database structure and complaint processing workflow based on feedback and updated the backend implementation strategy.
- Week 7: Compared the proposed backend architecture with existing systems and documented the advantages of the complaint processing and database design.
- Week 8: Reviewed the backend architecture, verified database relationships, and documented minor design corrections.
- Week 9: Demonstrated backend complaint processing, tested APIs and database connectivity, and verified complaint storage and retrieval.
- Week 10: Finalized backend development and database integration, tested authentication and complaint processing, resolved minor issues, and updated backend documentation.

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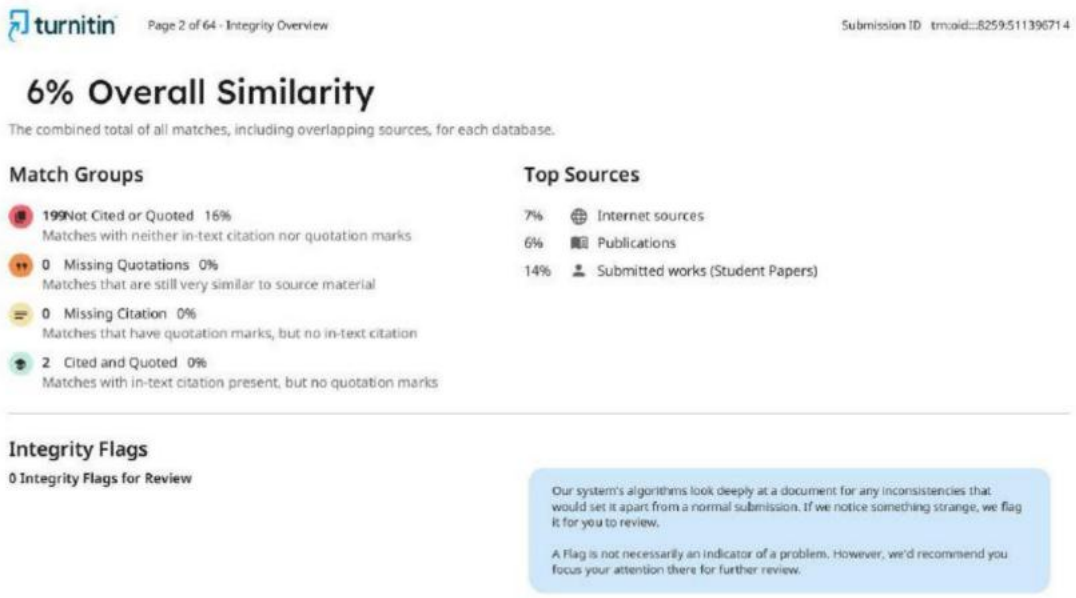
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